[Affiliation]

October 26, 2026

Editor-in-Chief
Advanced Engineering Informatics

Dear Editor,

Please find enclosed our manuscript "Coverage-controlled learning of connection-invariant magnetic surrogates for reconfigurable electric machines", which we submit for consideration as a research paper in Advanced Engineering Informatics.

Reconfigurable windings are an established route to wide-speed-range, high-efficiency electrified powertrains, yet every study we are aware of evaluates each candidate connection with its own dedicated FE model or flux-map table, and schedules mode changes with fixed thresholds or greedy rules. Our manuscript addresses both gaps with one idea: a winding connection is a set of linear constraints on coil currents, so a single connection-invariant surrogate—trained once on coil-level FE data—can query unseen connections; coverage determines when zero-shot transfer is reliable. We think this fits the journal's scope in three specific ways: (i) it is an engineering-informatics method (learned surrogates embedded in design and operation workflows) rather than a new machine or a new controller; (ii) it comes with a reproducible benchmark—7 hold-out connections, three difficulty tiers, 9,276+ nonlinear FE samples, fixed splits and protocols—released with code and data; and (iii) it delivers actionable engineering findings, including two data-collection design principles (what *kind* of current patterns to sample, which matters more than how many), a quantified physical-consistency advantage of co-energy-structured surrogates, an instructive negative result on monotonicity-constrained architectures, and a drive-cycle dynamic-program that prices reconfiguration honestly (changeover losses, dwell constraints). An independent FEMM cross-check is included; it validates the coarse inductance and nonlinear flux trends while exposing a zero-shot torque failure for an untrained tap connection, so the manuscript's claims are explicitly scoped to what the evidence supports.

All results in the manuscript are reproducible from the released code and data with one command per figure; the FE reference data, trained models, and benchmark splits are included. The work has not been published elsewhere and is not under consideration elsewhere. The authors declare no competing interests.

We suggest the following potential reviewers with relevant expertise: [3–6 suggested reviewers with affiliations and emails].

Thank you for your consideration.

Sincerely,

[Author Name]